Physical Chemistry Cumulative Exam

September 30, 2016

1. Describe the principle of Raman spectroscopy

2. Sketch the experimental arrangement of a Raman spectrometer

3. What are, if any, the limitations on the phase (solid, liquid, gas) of a sample for Raman spectroscopy?

4. The internal state of a molecule giving a Raman effect undergoes a change. What is the nature of this change for a typical Raman application? For instance, a change in the electronic state (orbital occupancy), a change in the vibrational state of the molecule, a change in the electron spin, or a change in the spins of the nuclei?

5. What is a typical source of electromagnetic radiation for Raman spectroscopy, and at what wavelength does a typical Raman spectrometer operate?

6. Describe the principle of microwave spectroscopy. Which property or properties of a molecule can be determined in microwave spectroscopy?

7. What is a typical wavelength for microwave spectroscopy?

8. Describe the limitations on the phase (solid, liquid, gas) of a sample for microwave spectroscopy.

9. What are the changes in the internal state of a molecule in microwave spectroscopy?

10. Electron paramagnetic resonance (EPR), also called electron spin resonance (ESR) is an important experimental method. Describe the principle of EPR.

11. Most substances don't have an EPR spectrum. What is the criterion for a molecule to have an EPR spectrum?

12. What information can be gained from the EPR spectrum?

All questions are worth maximum 5 points (0-5). Passing grade is 30 points or more.